

Scales of Measurement

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Introduction

Every variable in a dataset carries a scale of measurement, and the scale dictates which statistics are meaningful to compute. Stevens' 1946 taxonomy – nominal, ordinal, interval, ratio – remains the default vocabulary for this idea, and even when its strict version is criticised, the underlying distinctions are unavoidable in applied work. Computing a mean of jersey numbers is nonsense because jersey numbers are nominal; computing a median of exam grades is sensible because grades are at least ordinal.

Prerequisites

A reader should know what a variable is and should be able to tell numeric data from categorical data in R (as `numeric` vs. `factor` or `character`).

Theory

- **Nominal:** labels without order. Operations: equality. Examples: sex, blood type, country. Meaningful summaries: mode, frequency counts.
- **Ordinal:** ordered labels without meaningful distances. Operations: $=$, $<$, $>$. Examples: Likert scale, disease stage (I–IV), pain rating. Meaningful summaries: median, quantiles, rank correlations.
- **Interval:** ordered, with meaningful distances but no true zero. Operations: $+$, $-$. Examples: temperature in Celsius, calendar dates. Ratios are *not* meaningful: 40 °C is not “twice as hot” as 20 °C.
- **Ratio:** interval with a true zero. Operations: $\times$, $/$. Examples: mass, concentration, time elapsed, counts. Ratios are meaningful: 40 kg is twice 20 kg.

The scale determines permissible summary statistics and inferential procedures. A Pearson correlation requires interval or ratio data; a Spearman correlation extends to ordinal. A t-test operates on (at minimum) interval data, though in practice it is widely applied to Likert-like composites with reasonable robustness.

Assumptions

The taxonomy itself is descriptive rather than inferential; its use rests on the assumption that the chosen scale accurately characterises the variable. In practice, many variables sit ambiguously – is a 5-point Likert item ordinal or interval? Standard practice treats Likert *composite scales* (averages of many items) as interval and *single items* as ordinal.

R Implementation

```
library(ggplot2)

df <- data.frame(
  id      = 1:8,
  country = factor(c("DE", "US", "FR", "DE", "US", "FR", "DE", "US")),
  stage   = factor(c("I", "II", "III", "II", "IV", "I", "III", "II"),
                    levels = c("I", "II", "III", "IV"), ordered = TRUE),
  temp_c  = c(36.5, 37.1, 38.2, 36.9, 37.4, 36.7, 38.0, 37.2),
  mass_kg = c(72.1, 81.3, 65.8, 74.2, 88.7, 61.4, 79.0, 84.5)
```

```
)

table(df$country)
median(as.numeric(df$stage))
mean(df$temp_c)
mean(df$mass_kg); df$mass_kg[1] / df$mass_kg[3]
```

Nominal `country` is summarised by frequencies; ordinal `stage` by the median (the raw factor must be coerced to numeric for the median call); interval `temp_c` by the mean; ratio `mass_kg` by the mean and by ratios.

Output & Results

```
DE FR US
3  2  3

[1] 2      # median stage (II)
[1] 37.25  # mean temperature in C
[1] 75.875 # mean mass in kg
[1] 1.096  # mass ratio is meaningful for ratio data
```

Interpretation

A paper’s methods section should list each variable and its scale, because the scale justifies the summary statistic shown in the descriptive-statistics table. “Disease stage (ordinal) is summarised by the median (IQR); body mass (ratio) by the mean (SD).” This prevents a reader from being surprised by, say, a mean of a four-level ordinal variable dressed up as a continuous measurement.

Practical Tips

- Encode ordinal variables in R as `factor(..., ordered = TRUE)` so that methods that respect ordering (proportional-odds models, Spearman correlation) pick up the order automatically.
- Resist the urge to compute means of ordinal variables for reporting; a mean of disease stages is uninterpretable even when it produces a number.
- Do not confuse a *count* (ratio) with a *code* (nominal): “category = 3” is nominal even though stored as an integer.
- Be careful with variables that look ratio but are not: pH is actually a logarithmic transformation of concentration, so differences on the pH scale correspond to ratios of hydrogen-ion concentration.
- When a variable’s scale is ambiguous (Likert items), pre-register your analytical choice – ordinal or interval – so reviewers cannot object after the fact.

For Reviewers

What to look for in a paper using this method.

- Common misapplications.
- Diagnostics that should be reported but often aren’t.
- Red flags in tables and figures.
- What to verify.
- What an adequate Methods paragraph must contain.

See also — labs in this chapter

- Scientific process and research workflow
- R, RStudio, Quarto, and renv toolchain
- Data types, tidy data, accuracy and precision

- Import, joins, and missingness with dplyr